Co-Benefits Analysis - White Paper

1.1 Introduction

This section presents research-backed evidence on four co-benefit branches of urban tree canopy expansion: temperature reduction, rainfall interception, air filtering, and additional social, ecological, and economic co-benefits. Co-benefits are important in this case because urban tree canopy can address several urban challenges at once. In Chennai, the same intervention can reduce heat, lower runoff, improve air quality, and improve urban livability. Each finding is based on cited papers from our literature review or on peer-reviewed studies identified through systematic search. The research findings are general in scope. Each section also includes a Chennai-specific context area to show how the evidence applies locally. Findings are presented in shaded blocks, with the source cited in brackets. The section also draws on the i-Tree Planting tool (USDA Forest Service, 2024) to quantify co-benefits for the 10 priority intervention zones identified in our GIS analysis. The paper focuses on the 30% tree canopy target from the 3-30-300 framework and applies this target to Chennai, India.

1.2 Branch A: Temperature Reduction

1.2.1 How Trees Reduce Temperature

Tree canopy lowers urban temperatures through shading (blocking solar radiation from reaching surfaces) and evapotranspiration (releasing water vapor that cools surrounding air).

1.2.2 Research Findings: Canopy and Temperature

A 10% increase in tree canopy reduces urban air temperature by 0.8°C. A 30% increase lowers air temperature by up to 1.5°C. *(Ziter et al., 2025)*

Tree canopy cover is the dominant cooling factor, explaining 67% of the spatial variation in urban air temperature. *(Ziter et al., 2025)*

In Phoenix residential neighborhoods, increasing canopy from 10% to 25% produced up to 2.0°C average daytime cooling at the local scale. *(Middel et al., 2014)*

Cooling increases nonlinearly as canopy cover rises, with the strongest additional cooling occurring once canopy cover exceeds about 40%, especially at the city-block scale. *(Ziter et al., 2019)*

Urban tree canopy mitigates approximately 48.6% of the urban heat island effect globally on average. *(McDonald et al., 2025)*

1.2.3 Research Findings: Temperature and Health

Each 0.1 increase in NDVI (a measure of vegetation greenness) reduces heat-related mortality by 6-10% in low-GDP regions. (Ouria et al., 2025)

1.2.4 Chennai Context

Chennai's mean Land Surface Temperature rose from 29.4°C in 2000 to 33.6°C in 2020, a 4.2°C increase over two decades. Maximum and minimum temperatures are expected to increase by 2.9°C and 3.3°C respectively by the end of the century. *(TNGCC & ICLEI South Asia, 2026)*

The number of annual heatwave days in Chennai is projected to increase from 42 to 81 by 2050. Tamil Nadu's average annual heatwave days doubled from 15 to 30 between 2011 and 2019, and the state's maximum temperature has increased by 1.5°C since 1981. In 2023, the state reported 12 heatwave-related deaths. *(TNGCC & ICLEI South Asia, 2026)*

A clear negative correlation exists between LST and NDVI in Chennai: increased vegetation cover is associated with lower surface temperatures. The relationship between built-up density and temperature is quantified as: LST = 17.211 x NDBI + 38.051, confirming that surface temperature rises proportionally with built-up density. *(TNGCC & ICLEI South Asia, 2026)*

25% of Chennai's wards (50 wards) have a very high heat risk index. These wards account for 30% of the city's total population but cover only 12% of the total city area, indicating concentrated heat risk in dense neighborhoods. *(TNGCC & ICLEI South Asia, 2026)*

Chennai's per capita green cover is 8.75 sq.m, below the WHO standard of 9.5 sq.m per person. The city's vegetation declined by 22% over the last two decades. Very high-risk wards have NDVI as low as 0.1 to 0.12, indicating minimal vegetation. *(TNGCC & ICLEI South Asia, 2026)*

Our team's SUHII analysis measured Chennai's urban heat island intensity at 5.024°C using the Halo method with a 10 km exclusion zone and C-Mask for coastal bias correction.

Built-up land in Chennai's periphery surged from 11.9% to 44.16%, intensifying heat exposure. *(Muthiah et al., 2022)*

For India, the recommended social discount rate is 8% for general projects and 6% for long-life environmental and infrastructure projects. *(IEG/NITI Aayog, 2023)*

1.3 Branch B: Rainfall Interception

1.3.1 How Trees Reduce Flooding

Tree canopy intercepts rainfall before it reaches the ground. Leaves and branches catch water that then evaporates instead of becoming surface runoff. Root systems promote soil infiltration. Both reduce stormwater volume entering drainage systems. The same canopy expansion that cools a neighborhood also reduces flood risk.

1.3.2 Research Findings: How Much Rainfall Do Trees Intercept?

Individual urban trees intercept between 15% and 28% of rainfall falling on their canopy area. Depending on species, canopies can retain between 0.03 and 2.24 mm of water per square meter of leaf area. *(Kuehler et al., 2017; Center for Watershed Protection, 2016)*

In closed-canopy forests, interception loss is approximately 18-29% of total precipitation for hardwood forests, and approximately 18-45% for coniferous forests. *(Carlyle-Moses & Gash, 2011)*

Broadleaf evergreen and conifer stands intercept 36% of summer rainfall (leaf area index = 6.1). Mixed conifer and deciduous stands intercept 18% (leaf area index = 3.7). *(Xiao & McPherson, 1998)*

Individual street trees intercept 610 to 26,000 liters of annual rainfall per tree, varying by species and tree characteristics. *(Xiao & McPherson, 2002)*

Urban birch trees intercept 25.6% of gross rainfall on average. Canopy attenuates average rainfall intensity by 50.2% during the leafed season and 41.6% during the leafless season. *(Ljubljana Canopy Study, 2023)*

1.3.3 Research Findings: How Much Does This Reduce Runoff?

Urban trees can reduce stormwater runoff volume by 20-25% and peak flow by 16-25%. Infiltration and storage beneath tree canopies account for over 20% of the water balance. *(Alivio et al., 2025)*

Removing 31 mature street trees from a residential catchment increased surface runoff by 198 cubic meters (4% of total runoff). Per tree, 66 liters of rainfall per square meter of canopy was lost that would otherwise have been intercepted. *(Selbig et al., 2022)*

Even during intense, high-volume storm events, trees still provide up to 13.2% reduction in runoff. Mixed-species plantings enhance hydrological performance compared to single-species planting. *(Alivio et al., 2025)*

1.3.4 Chennai Flood Context

Chennai receives approximately 1,300 mm of rainfall annually. Recent years have seen intense downpours over short periods leading to urban flooding. Annual rainfall is expected to decrease by 5% by 2050, but the intensity of individual events is increasing. *(TNGCC & ICLEI South Asia, 2026)*

Wards 32, 44, 76, and 78 in Chennai are identified for bioswales, rain gardens, and waterbody restoration because they face both high heat risk and frequent waterlogging. These wards have slum settlements along the Buckingham Canal. The report states that restoring canal edges will 'mitigate both heat and flood risk for these vulnerable populations.' *(TNGCC & ICLEI South Asia, 2026)*

Urban growth and sprawl in Chennai directly increase flood depth, extent, and inundation area under future development scenarios. The same urbanization that intensifies the heat island also worsens flooding. *(Nithila Devi et al., 2019)*

When stormwater co-benefits are excluded from CBA, grey infrastructure dominates. When included, green-grey and green-blue-grey portfolios become economically viable. NBS strategies achieved BCR greater than 1 in 3 of 4 case study cities over a 30-year life cycle. *(Alves et al., 2019; Ruangpan et al., 2024)*

1.4 Branch C: Air Filtering

1.4.1 How Trees Clean Air

Tree leaves physically capture particulate matter (PM2.5, PM10) on their surfaces. They absorb gaseous pollutants (NO2, SO2, O3) through stomata. Once inside the leaf, gases diffuse into intercellular spaces and are permanently converted. Trees also sequester carbon dioxide, storing it as biomass.

1.4.2 Research Findings: How Much Pollution Do Trees Remove?

Urban trees remove an average of 6.73 grams of pollutants per square meter of canopy cover per year across all pollutant types (NO2, O3, PM2.5, SO2). *(Nowak et al., 2014)*

In Strasbourg, France, city-managed trees removed approximately 88 tonnes of pollutants in one year: 1 tonne of CO, 14 tonnes of NO2, 56 tonnes of O3, 12 tonnes of PM10, 5 tonnes of PM2.5, and 1 tonne of SO2. Urban trees reduced about 7% of emitted PM10 in the city's atmosphere. *(Selmi et al., 2023)*

Across the United States, urban trees remove an estimated 711,000 tonnes of particulate matter per year. *(Nowak et al., cited in PMC, 2023)*

In Guangzhou, China, urban vegetation removes 312 megagrams (metric tonnes) of SO2, NO2, and total suspended particles annually. *(Guangzhou Study, cited in PMC, 2023)*

1.4.3 Research Findings: Health Impact of Air Pollution Removal

PM2.5 removal by urban trees ranged from 4.7 tonnes per year in Syracuse to 64.5 tonnes per year in Atlanta. The monetary value of this removal ranged from $1.1 million per year (Syracuse) to $60.1 million per year (New York City). Most of this value came from reducing human mortality. *(Nowak et al., 2013)*

Trees reduce heat-related mortality by approximately 1 person per year per city on average, and up to 7.6 people per year in New York City, through PM2.5 removal alone. *(Nowak et al., 2013)*

A 10 micrograms per cubic meter increase in PM2.5 is associated with approximately 4% increased risk in all-cause mortality, 6% in cardiopulmonary mortality, and 8% in lung cancer mortality. *(Pope et al., 2002)*

1.4.4 Chennai Context

Energy consumption in Chennai's buildings sector contributed 71% of total GHG emissions, mainly due to reliance on thermal power for urban cooling. This highlights the need to incorporate heat resilience and energy-efficient cooling into built infrastructure. *(TNGCC & ICLEI South Asia, 2026)*

Street tree planting can reduce cooling loads in adjacent buildings by 5-10%. Vertical gardens and green walls can reduce building cooling loads by 5-15%. *(Chennai Co-Benefits Analysis Matrix, 2026)*

Chennai faces serious air quality challenges, particularly with PM2.5 and NO2. Tree canopy expansion in priority tessellations would concentrate filtration benefits in neighborhoods with the highest combined heat exposure and social vulnerability. The i-Tree Planting tool quantifies pollutant removal as grams of PM2.5, O3, NO2, and SO2 removed per year, plus kg of CO2 stored, which can be re-priced using India-specific health cost estimates.

1.5 Branch D: Additional Co-Benefits

Beyond temperature reduction, rainfall interception, and air filtering, urban tree canopy expansion generates several additional co-benefits that don't fit neatly into the first three branches but are documented in the literature and relevant to Chennai.

1.5.1 Public Health and Mental Wellbeing

Tree planting in high-risk Chennai wards (54-57) directly benefits slum communities along riverbanks and canals who are highly vulnerable to urban heat. Pocket parks and community gardens provide heat relief, mental health benefits, community cohesion, and opportunities for physical activity. *(Chennai Co-Benefits Analysis Matrix, 2026; TNGCC & ICLEI South Asia, 2026)*

Wards 44, 48, 52, 55, 73, and 75 together contain 45 schools, 11 markets, and 35 bus stops. Shading structures and tree planting in these public spaces would improve thermal comfort for commuters, pedestrians, and children. *(TNGCC & ICLEI South Asia, 2026)*

14 wards in Chennai have very high vulnerability due to higher slum populations, inadequate road infrastructure, low NDVI, fewer parks, and limited medical facilities. Areas like Nochikuppam and Ayothikuppam have less than 1 park per sq.km compared to the city average of about 1.5. *(TNGCC & ICLEI South Asia, 2026)*

1.5.2 Biodiversity and Ecology

Street tree planting and micro forests create habitat for urban birds and insects, improving urban biodiversity. The Tamil Nadu government's Karunkadugal initiative promotes development of mini forests in cities using dense, multi-species native plantations. *(TNGCC & ICLEI South Asia, 2026)*

Waterbody restoration in wards 32, 44, 76, and 78 supports wetland habitat restoration and aquatic biodiversity alongside heat and flood risk reduction. *(Chennai Co-Benefits Analysis Matrix, 2026; TNGCC & ICLEI South Asia, 2026)*

1.5.3 Economic Benefits and Employment

Tree planting and urban greening create jobs in nurseries, planting and maintenance, urban forestry, ecological restoration, and park management. MSME clusters in Chennai (wards 32, 44, 48, 51, 54, 56, 57, 63, 73, 75, 104, 109, 105, 130, 168) can benefit from cooling interventions including landscaping, energy-efficient cooling appliances, and solar rooftops. *(Chennai Co-Benefits Analysis Matrix, 2026; TNGCC & ICLEI South Asia, 2026)*

Cool roofs can reduce building cooling energy demand by 10-30%. This is particularly relevant in Chennai where building energy consumption accounts for 71% of total GHG emissions. *(Chennai Co-Benefits Analysis Matrix, 2026; TNGCC & ICLEI South Asia, 2026)*

1.6 Connecting the Four Branches

1.6.1 Why Co-Benefits Must Be Included in Appraisal

Tree planting programs are often seen as weak investments. When cities evaluate them purely on cooling or purely on aesthetics, the costs of planting and decades of maintenance can outweigh the measured returns. This leads decision-makers to favor grey infrastructure like concrete drainage or engineered cooling systems, which show clearer single-purpose returns.

But this framing is flawed. It measures one benefit and ignores the rest. As this paper has shown, the same trees that reduce temperature also intercept rainfall, filter air pollution, store carbon, improve public health, and support biodiversity. When appraisals only count one of these, they systematically undervalue green coverage.

The research confirms this:

When co-benefits are excluded from cost-benefit analysis, grey infrastructure dominates the feasibility rankings. When co-benefits are included (energy savings, air pollution reduction, carbon sequestration, thermal comfort valued through willingness-to-pay), combined green-grey and green-blue-grey solutions become economically viable. The authors note the co-benefits counted were conservative. *(Alves et al., 2019)*

> NBS strategies achieved a Benefit-Cost Ratio above 1 in three of four case study cities over a 30-year life cycle. Even in the fourth city where BCR was below 1, NBS was still more economically efficient than grey infrastructure. *(Ruangpan et al., 2024)*
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> CBA outcomes depend strongly on which benefits are counted. Studies that include a broader set of co-benefits consistently find stronger results for green infrastructure. Studies with a narrow scope favor grey. Biological maturation time of NBS is rarely incorporated even though it can materially affect benefit streams.*(Chelli et al., 2024)*

This is the ranking flip. Tree planting doesn't become a better investment when co-benefits are added. It was always that good. Traditional appraisal just wasn't counting properly.

1.6.2 Recommended Frameworks for Chennai

Several established frameworks exist for evaluating co-benefits. Each takes a different approach depending on what needs to be measured and who the audience is.

> Life-Cycle Cost-Benefit Analysis (LCCBA) lays out all benefits and costs year by year over the project's lifetime, discounts future values to the present, and produces standard investment metrics like NPV and BCR. Ruangpan et al. used this approach with a 30-year horizon and a 3% discount rate across four cities, evaluating four NBS measures and five co-benefit categories. *(Ruangpan et al., 2024)*
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> Multi-Criteria Decision Analysis (MCDA) scores interventions across multiple objectives and lets stakeholders assign weights based on local priorities. Alves et al. used this to rank green and grey infrastructure across flood reduction, cost, and co-benefits using a 1-to-5 scoring scale, with preference weights from 1-10 for five co-benefit sub-categories. Robustness was tested with 5,000 Monte Carlo simulations. *(Alves et al., 2018)*
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> InVEST (Integrated Valuation of Ecosystem Services and Tradeoffs) models how much ecosystem service an intervention generates under different scenarios. Biasin et al. used it in Turin to evaluate NBS scenarios addressing both UHI and flooding simultaneously as joint climate risks. *(Biasin et al., 2023)*

For Chennai, we recommend MCDA as the primary framework, because it can accommodate both monetizable benefits (energy savings, avoided flood damage) and the harder-to-quantify benefits (equity, biodiversity, public health) that matter to the World Bank and local stakeholders.

As a practical extension, we are using the **i-Tree Planting tool** to generate the physical co-benefit quantities (stormwater intercepted, CO_2 stored, pollutants removed, energy saved) for priority zones in our GIS analysis. These i-Tree outputs feed into the broader MCDA evaluation as quantitative inputs, re-priced for Chennai's local costs.

1.7 Summary of Key Findings

Branch	Finding	Metric	Source
A	10% canopy increase reduces air temp by 0.8°C	0.8°C per 10%	Ziter et al., 2025
A	30% canopy increase reduces air temp by up to 1.5°C	1.5°C per 30%	Ziter et al., 2025
A	Each 0.1 NDVI increase reduces heat mortality 6-10% in low-GDP regions	6-10% per 0.1 NDVI	Ouria et al., 2025
A	Energy savings quantified as €/house/year over 30-year horizon	€/house/year	Alves et al., 2019
B	Trees intercept 15-28% of rainfall; broadleaf evergreens intercept up to 36%	15-36% interception	Kuehler et al., 2017; Xiao & McPherson, 1998
B	Urban trees reduce stormwater runoff volume by 20-25% and peak flow by 16-25%	20-25% runoff reduction	Alivio et al., 2025
B	Individual trees intercept 610 to 26,000 liters of rainfall per year	610-26,000 L/tree/yr	Xiao & McPherson, 2002
C	Urban trees remove 6.73 g of pollutants per m² canopy per year	6.73 g/m²/yr	Nowak et al., 2014
C	PM2.5 removal: 4.7-64.5 tonnes/yr per city, valued $1.1M-$60.1M/yr	$1.1M-$60.1M/yr	Nowak et al., 2013
C	Strasbourg trees removed 88t pollutants/yr including 56t O3, 14t NO2, 5t PM2.5	88 tonnes/yr	Selmi et al., 2013
A	Chennai LST rose from 29.4°C to 33.6°C (2000-2020); heatwave days to double by 2050	4.2°C increase	TNGCC & ICLEI South Asia, 2026
A	25% of Chennai wards (50) have very high heat risk, covering 30% of population	50 wards / 30% pop	TNGCC & ICLEI South Asia, 2026
B	Wards 32,44,76,78 face both heat and flood risk; canal restoration mitigates both	Dual-risk wards	TNGCC & ICLEI South Asia, 2026

C	Chennai building energy = 71% of GHG emissions; trees reduce cooling loads 5-10%	71% emissions; 5-10%	TNGCC & ICLEI South Asia, 2026
D	Cool roofs reduce cooling energy demand by 10-30% in buildings	10-30% savings	Chennai Co-Benefits Matrix, 2026
D	Priority wards have <1 park/sq.km vs city average 1.5; per capita green 8.75 vs WHO 9.5	Green space deficit	TNGCC & ICLEI South Asia, 2026
All	Standard CBA undervalues benefits to lower-income communities	Equity gap	Environment & Development Economics, 2021
All	India discount rate: 6% for environmental infrastructure	6% base case	IEG/NITI Aayog, 2023

1.8 i-Tree Planting Tool: Quantifying Co-Benefits for Chennai

1.8.1 What is the i-Tree Planting Tool?

i-Tree Planting is a free web-based tool developed by the USDA Forest Service that estimates the environmental co-benefits of tree planting projects over a defined time horizon. Given inputs including species, tree size at planting, location, and project duration, the tool forecasts cumulative benefits across five categories: carbon sequestration, stormwater interception, avoided runoff, air pollutant removal, and building energy savings (USDA Forest Service, 2024). For this project, i-Tree Planting was used to generate physical co-benefit quantities for each of the 10 priority hexagon zones identified in the GIS analysis. A full methodological account is available in the standalone co-benefits analysis white paper.

1.8.2 Configuration for Chennai

Since Chennai is not available as a location in i-Tree, US National Average was selected following standard practice for non-US projects. Key parameters were localised as follows:

Parameter	Value	Source
Electricity Emissions Factor	710 kg CO2 eq/MWh	CEA, 2024
Fuel Emissions Factor	63 kg CO2 eq/MMBtu	IPCC, 2006
Project Duration	40 years	Standard practice

Tree Mortality	3% per year (Annual)	Hilbert et al., 2019
Species	Albizia saman (Rain Tree)	Selected for Chennai

Albizia saman was selected for its large mature canopy spread of 15 to 25 metres, fast growth rate, and proven tolerance of Chennai's tropical heat, humidity, and monsoon conditions.

1.8.3 Number of Trees and Group Structure

The number of trees required per zone was calculated from each hexagon's canopy gap relative to the 30% target, applied over a hexagon area of 2 km2, with 200 m2 assigned as the conservative mature canopy area per Albizia saman. Five zones with 0% existing canopy were grouped together in i-Tree, resulting in 6 input groups representing all 10 zones. Total trees across all 10 zones: **29,778**.

1.8.4 Results and Interpretation

Over the 40-year project period, the 10 priority zones are projected to deliver the following co-benefits:

Co-Benefit	Total Across 10 Zones
CO2 Sequestered	3,397 tonnes
Avoided Stormwater Runoff	49,307 m3
Rainfall Interception	~860,000 m3
O3 Removed	~74,000 kg
NO2 Removed	~6,150 kg

SO2 Removed	~13,270 kg
PM2.5 Removed	~3,890 kg

Carbon sequestration of 3,397 tonnes over 40 years reflects the slow but sustained biomass accumulation of Albizia saman at scale. At an average of around 340 tonnes per zone, this is consistent with Nowak et al.'s (2014) estimate of 6.73 grams of pollutants removed per square metre of canopy per year when applied across nearly 30,000 trees over four decades. These sequestration values should be read alongside the broader carbon context in Section 8.4, where building energy consumption accounts for 71% of Chennai's total GHG emissions (TNGCC & ICLEI South Asia, 2026). Tree planting alone will not close that gap, but it contributes meaningfully and delivers co-benefits that energy efficiency measures do not.

Stormwater avoided runoff of 49,307 m3 is particularly relevant to Chennai's flood risk profile. This aligns with Branch B findings that urban trees reduce runoff volume by 20 to 25% (Alivio et al., 2025), and is especially significant for wards 32, 44, 76, and 78, which face both high heat risk and frequent waterlogging (TNGCC & ICLEI South Asia, 2026).

PM2.5 removal carries the greatest health significance of the air quality results. As noted in Section 8.4, a 10 microgram per cubic metre increase in PM2.5 is associated with a 4% increase in all-cause mortality (Pope et al., 2002). The priority zones were selected in part for their high social vulnerability scores, meaning these gains will benefit residents least able to protect themselves from pollution exposure.

Energy savings returned zero across all groups due to the tool's US-based location setting and the exclusion of building climate controls, which do not apply to Chennai's tropical context. This benefit category should be estimated separately using Chennai-specific building energy models.

A more detailed breakdown of results by group and zone, including the methodology and limitations, is provided in the standalone co-benefits analysis conducted using the i-Tree tool.

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